

Polar and Dual-Polar Schemes: Permutation Formula Proof and Exact Verification for $Sp_4(2)$

Executable companion note

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Abstract

We prove the fixed-polar-subspace version of the permutation Molien formula and test two nontrivial actions of the explicitly constructed matrix group $Sp_4(2)$: isotropic points and maximal totally isotropic 2-spaces. The latter is a rank-two dual-polar scheme.

1 Cycle and orbit formulas

Let a finite isometry group G act on a finite family X of totally isotropic or singular subspaces, and put $V = \mathbb{C}[X]$. If $c_\ell(g)$ counts ℓ -cycles on X , the determinant of an ℓ -cycle block is $1 - t^\ell$, so

$$\mathcal{S}_{G,X}(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{i,\ell} (1 - t_i^\ell)^{-c_\ell(g)}. \quad (1)$$

Since the monomial basis of $\text{Sym}^d(V)$ is indexed by d -multisets in X , Burnside's lemma independently gives

$$[m_\tau] \mathcal{S}_{G,X} = \# \left(G \backslash \prod_j \text{Sym}^{\tau_j}(X) \right). \quad (2)$$

Define the geometric fixed-point count

$$F_r^{\text{pol}}(g) = \#\{U \in X : g^r U = U\}. \quad (3)$$

The universal relation $F_r = \sum_{\ell|r} \ell c_\ell$ and Möbius inversion yield

$$c_\ell^{\text{pol}}(g) = \frac{1}{\ell} \sum_{d|\ell} \mu(\ell/d) F_d^{\text{pol}}(g). \quad (4)$$

Equations (1)–(4) prove the proposed formula. Relative to the Grassmann case, the only additional predicate in (3) is total isotropy or singularity.

For maximal isotropic subspaces in Witt rank r , pair orbits are classified by $r - \dim(U \cap W)$. Consequently

$$[m_{(1)}] = 1, \quad [m_{(2)}] = [m_{(1,1)}] = r + 1.$$

2 Concrete symplectic construction

On $\mathbb{F}_2^4$ use the nondegenerate alternating form

$$\langle x, y \rangle = x_1y_2 + x_2y_1 + x_3y_4 + x_4y_3.$$

The code enumerates all 20,160 matrices in $GL_4(2)$ and retains exactly the 720 satisfying $\langle gx, gy \rangle = \langle x, y \rangle$ on basis vectors. It separately enumerates reduced-row-echelon subspaces. There are 15 one-spaces (all isotropic for an alternating form) and 15 two-spaces on which the form vanishes identically. Thus both induced representations have dimension 15, but their vertices are different geometric objects.

For every group matrix and every needed power, the program acts on the stored bases and counts invariant members of X directly. Equation (4) must reproduce the cycles of the separately constructed vertex permutation.

3 Three independent coefficient routes

For the actual 15×15 permutation matrix P , complete symmetric characters are computed without cycle data from

$$dh_d(P) = \sum_{a=1}^d \text{tr}(P^a) h_{d-a}(P), \quad h_0 = 1.$$

The matrix-group target is the average of $\prod_j h_{\tau_j}(P)$. A second calculation extracts the coefficient from (1); a third explicitly enumerates the configuration orbits in (2). Agreement of these routes tests the representation, the proposed formula, and its geometric interpretation separately.

4 Exact results

Values are ordered by $\tau = (1), (2), (1, 1), (3), (2, 1), (1, 1, 1)$.

Action	$ G $	$ X $	common coefficient vector
$Sp_4(2)$ on isotropic points	720	15	(1, 3, 3, 8, 12, 16)
$Sp_4(2)$ on maximal isotropic 2-spaces	720	15	(1, 3, 3, 8, 12, 16)

All 12 matrix/cycle/orbit comparisons and all geometric fixed-point reconstructions pass. The dual-polar pair value is $3 = r + 1$ for $r = 2$. For the maximal action at $(t_1, t_2) = (0.037, 0.061)$, both numerical averages equal 1.1263214575673286 (reported error 0).

5 Scope and reproduction

The exact proof applies equally to alternating, Hermitian, and quadratic forms. The executable representative is symplectic because $Sp_4(2)$ is large enough to exhibit all rank-two pair relations while remaining fully enumerable and auditable.

Run

```
marimo run for_this_guy/association_scheme_permutation_representations/
    polar/notebook.py
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from the repository root to inspect every comparison.